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Comparative Analysis of Multimodal Emotion Detection Approaches

Bachelor thesis

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Abstract

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Contents

1. Introduction **1**
1.1. Motivation . . . 1
1.2. Structure . . . 1

2. Background **1**
2.1. Emotion . . . 2
2.1.1. Emotion Classification . . . 3
2.1.2. Emotion in Text . . . 4
2.1.3. Emotion in Speech . . . 5
2.1.4. Emotion in Facial Expression . . . 5
2.2. Classification Approaches . . . 6
2.2.1. Classification in Text . . . 6
2.2.2. Classification in Audio . . . 10
2.2.3. Classification in Images and Videos . . . 10
2.3. Datasets . . . 11

Bibliography **14**

Chapter 1.

Introduction

1.1. Motivation

motivation

1.2. Structure

structure

Chapter 2.

Background

This chapter will give an introduction and overview on different approaches to classify types of emotions with respect to their advantages and disadvantages. Furthermore it will explain the approaches to recognize emotion in the mediums text, video and audio. Based on this common implementation techniques for supervised and unsupervised approaches are presented with a focus on convolution and transformer models.

2.1. Emotion

There is no unambigious definition for the term ‘emotion’. There are many different emotions, with different level of intensity. Considering the question of what an emotion is and where it comes from, William James discussed two different theories about the origin of them. Either there are seperate areas in the cerebrum or else they correspond to motory and sensory processes [1]. Many work in psychology add further aspects to the latter. In order to provide a framework, the following chapters will focus just on the given emotion and how to recognize it.

explain more the difference

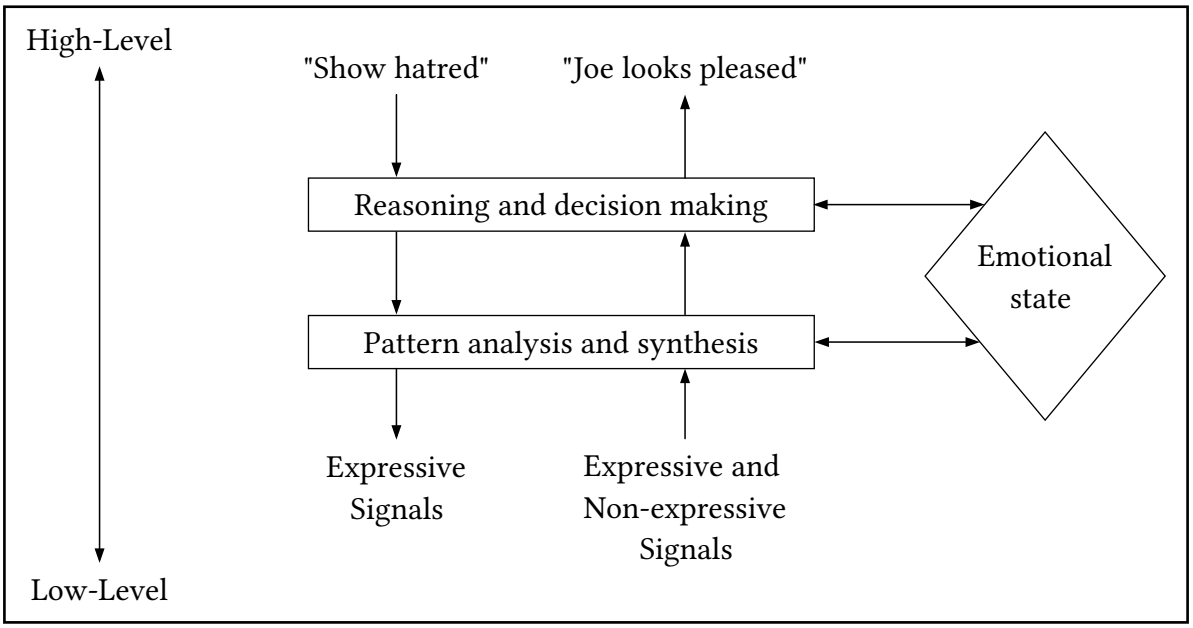


Figure 2.1.: Information flows from both high to low to high in a system that can express, recognize and “have” emotions [2].

Figure 2.1 presents that information like motory, sensory processes, their combination and the differentiation from semantic and syntatic context are categorized in high-level and low level information to process an emotion.

High-level information are defined by the semantic of a situation which can be drawn from audiovisual processes for example the command “Show hatred”. To show hate a certain emotional state has to be reached and must be reasoned by the situation. As a conclusion low-level information, for instance face expressions, gestures or other syntactic articulations can be observed from the person or system wich executes the given instruction. These patterns are analyzed or learned and synthesized by the process before.

On the other direction, these low-level expressive information can be combined to get the semantic of the emotional state. In addition there can be non-expressive data as a valid input. It describes the passive influence on a persons state like the weather or the setting of the environment. Most datasets won’t cover this type of information.

The mid-level builds patterns for the given input and combines different low-level signals to synthesize better reasioning for a high-level output. For example a slightly faster speech rate and a tense articulation can be clustered together to get classified as an pleased look.

2.1.1. Emotion Classification

There are different apporaches to classify the semantic high-level features. The most popular classification is the basic-emotion framework by Paul Ekman [3] describing six different emotions: anger, disgust, fear, surprise, sadness and happiness. This arose from different facial expression, which Ekman categorized to the corresponding emotion.

Over the time different frameworks were built on this foundation to introduce for each basic emotion multiple detailled emotions. A common model is the Plutchik wheel of emotion, which is commonly used to classify emotion in text. It offers eight types of basic emotion: joy, trust, fear, surprise and their opposites sadness, disgust, anger and anticipation [4].

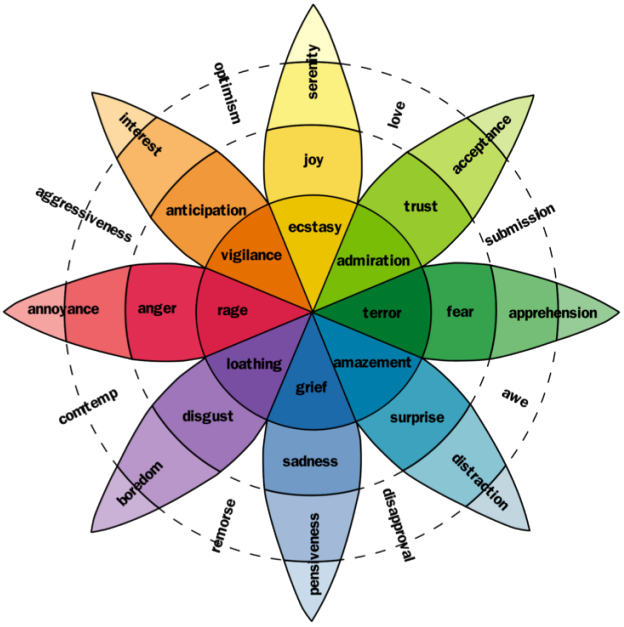


Figure 2.2.: Plutchik Wheel of Emotion [5].

Visible in Figure 2.2 each emotion has multiple levels of intensity. For example, the enhanced level for disgust is loathing, a more moderate level is boredom. In total there are 24 different emotions.

Why can it be better to have more emotion variety?

Although there will be a loss of detail, the content of this thesis will focus on the basic emotion from Paul Ekman. The wheel of emotion from Plutchik is suitable for emotion detection in text but most datasets in facial emotion recognition are focused on Ekmans emotion theory .

Why I wont use the Wheel of Emotion

2.1.2. Emotion in Text

The most common approach to classify emotion from natural language text is to classify the semantic content from these sentences. In order to make it possible, two syntactical assumptions are needed.

1. There is a function that is able to map words and sentences to their corresponding emotion.

This can be approached by training a deep neural network with a suitable dataset or a clustering algorithm that measures the similarity between words and sentences to match their corresponding emotion label. The function must be surjective, so a token won't be able to be labeled with more than one emotion at the time.

For a relieable result in training, the second assumption is important.

2. Words and sentences do not change their meaning over time.

This is also the flaw from emotion detection in text. For example the sentence "I'm glad to be here"

briefly explain irony which can lead to different meanings for the same sentence

2.1.3. Emotion in Speech

show different chanings in audio diagrams

Emotion / Feature	Fear	Anger	Sadness	Hapiness	Disgust
Speech Rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch Average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch Range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice Quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grubled chest tone
Pitch Changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2.1.: Features of an audio with correlating emotion [2], [6].

2.1.4. Emotion in Facial Expression

talk about the facs system

2.2. Classification Approaches

Over time many approaches to detect and classify emotion has been developed. There are severeral supervised architectures and unsupervised techniques which represent the current state of the art for emotion detection in text, audio and video.

2.2.1. Classification in Text

Word Embedding and Tokenization

The most important part of all classification approaches is called word embedding. It represents words as real-valued vectors [7]. It uses syntactic but also the semantic meaning of the words to get a useful vector. This technique is commonly used in natural language processing like information retrieval in text.

To generate such an embedding function A, a one-hot vector [8] is needed. It represents every word in a space of words as an different vector. For example the sentence:

$$\textit{The tree is green.}$$

will be first tokenized into a set of words V.

$$V = \{\textit{the}, \textit{tree}, \textit{is}, \textit{green}\}$$

The size of the set $|V| = 4$. As a result a mapping $w \in V$ to a $|V|$-dimensional vector is needed.

$$A : V \to \mathbb{R}^{|V|}$$

$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$
the	$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$	tree	$\begin{pmatrix}0\\1\\0\\0\end{pmatrix}$
is	$\begin{pmatrix}0\\0\\1\\0\end{pmatrix}$	green	$\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$

Table 2.2.: Possible one-hot vector mapping for the sentence “The tree is green.”

To measure the similarity the cosine similarity is often used [8], [9]. Similar words or synonyms should have a smaller angle between each other, than unsimilar words.

$$\text{sim}(w_1, w_2) = \cos(\theta) = \frac{w_1 \cdot w_2}{\|w_1\| \, \|w_2\|}$$

While the one-hot vector embeds it syntactically only, there won’t be any similarity or dissimilarity between the words. Therefore a semantic consideration will be useful.

The Continous Bag-of-Words Model, shortly CBOW is capable to take context of a word as an input to calculate an embedded vector with emphasis on syntax and semantic. It consists of a neural network which mainly constitutes of three layers: the input layer, a hidden layer and the output layer [10].

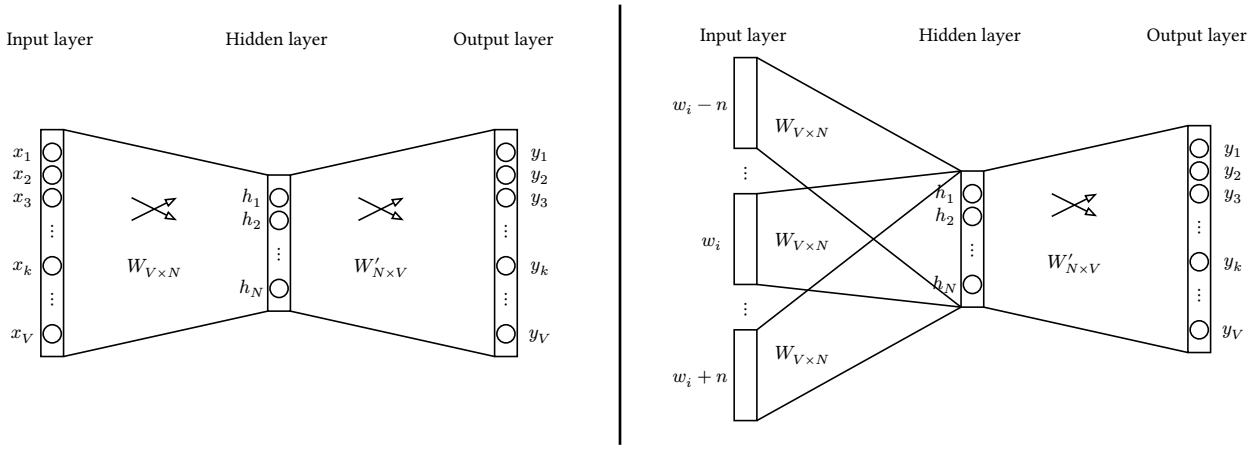


Figure 2.3.: Left: Simple CBOW-architecture for a single word; Right: CBOW-architecture for multiple words. Words have length k and are padded to size V to match the dimension [8].

The architecture shown on the right side of Figure 2.3 takes one one-hot vector $A(w_i)$ as an input expression. To get the context of the inputted term, phrases before and after are needed. Therefore an addition to the architecture is necessary.

With unspuervised learning the model on the right of Figure 3 is able to generate a new vector which represents the context of a word. Now, under consideration of the cosine similarity, words with a similar context have a quite smaller angle then without any connection in semantics. In both architectures, the hidden layer acts like an arithmetic operator which combines the context to the outputted one-hot vector.

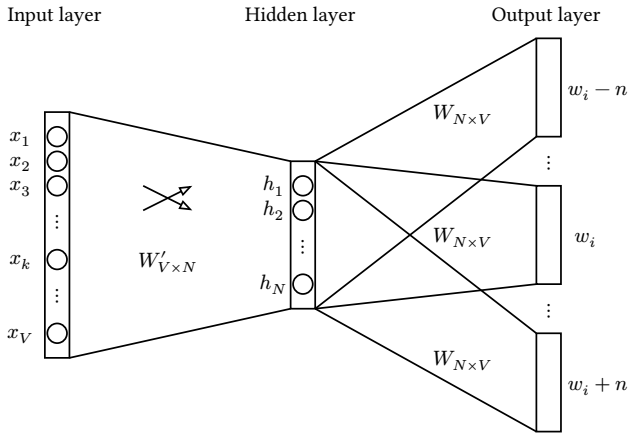


Figure 2.4.: Skip-Gram Model. All words have size V (after padding) [11].

The Skip-Gram Model calculates an embedding in reverse from the idea of the CBOW-Model. As the architecture shows in Figure 2.4 it tries to calculate the surrounding words with a given target term. Therefore a training like the CBOW with contextual phrases is needed before it can work like a semantical decoder. Both architectures are commonly used in different pretrained word embedding networks, like Word2Vec from Google [11], [12].

Fast WordPiece Tokenization is an approach to tokenize sentences and words in an efficient way by splitting words into smaller subwords. It is used in a language understanding transformer model from Google which is called Bert [13].

The basic problem from other tokenization methods like the normal WordPiece Tokenization starts in the vocabulary tokenization in tries. Trie memory is a paradigma which is able to store strings efficiently where each edge of the data structure stores phonemes. Traversing a path consisting of multiple edges results in a word. As a result the structure does not have to store reduntant parts of words [14].

The tokenization has to search for the longest token at a position. Therefore it starts from the shortest substring and iterates over the input text, while trie matching to match the longest prefix which matches a vocabulary token [15]. If the trie matching fails at the next character, the algorithm has to involve backtracking because the longest vocabulary token may be several characters back. This results into a time complexity of $O(nm)$ where m is the maximum vocabulary token length and n the input length.

Fast WordPiece Tokenization introduces so called failure links and pops. Failure pops are a list of longest-matching prefix tokens of a node in the trie, which can be removed in order until a failure link node is reached. The failure node is able to represent the missing suffix. This whole algorithm called LinMaxMatch has a time complexity of only $O(n)$.

Supervised Approaches

With a given word embedding and tokenization supervised learning approaches are able to recognize emotion in text. Therefore an annotated dataset is needed, which has a relieable mapping from sentence or word to the corresponding emotion.

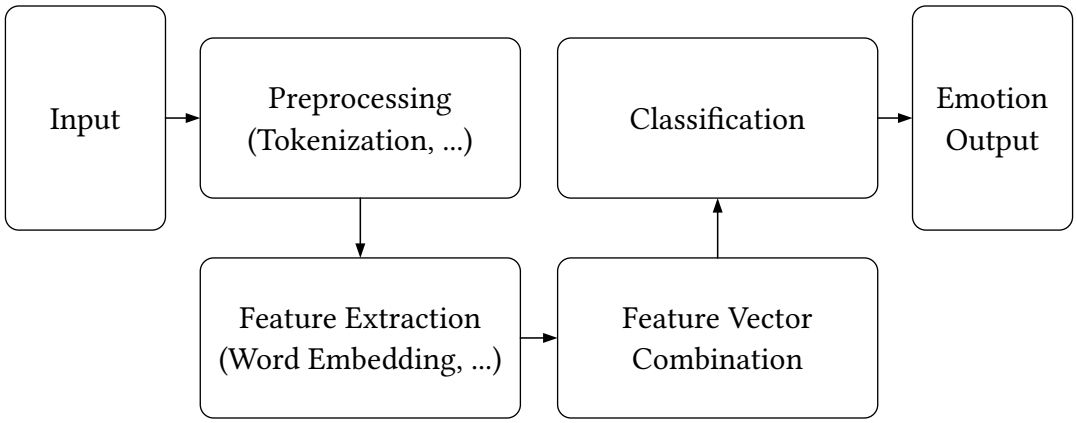


Figure 2.5.: Deep Learning Assisted Semantic Text Analysis

Most models are using the architecture paradigma shown in Figure 2.5 . They mostly differ in the step of combining the feature vectors ant the type of classification because it offers the biggest room to maneuver. A preprocessing and feature extraction is always necessary. Sometimes a WordNet [16] is integrated to get a better overview about synonyms or to paraphrase words.

Convolutional and Recurrent Neural Networks (CNN/RNN) are an effective approach on sentiment analysis of text [17]. Convulation consists of a kernel-funciton which downscales the input but keeps the important features. This technique is called Pooling and brings several benefits like less memory usage, less overfitting in training and a faster computation time. Herefore multiple words or sentences of the space V are combined to region matrices. On each region L convolutional filters are applied to lern local n-gram features.

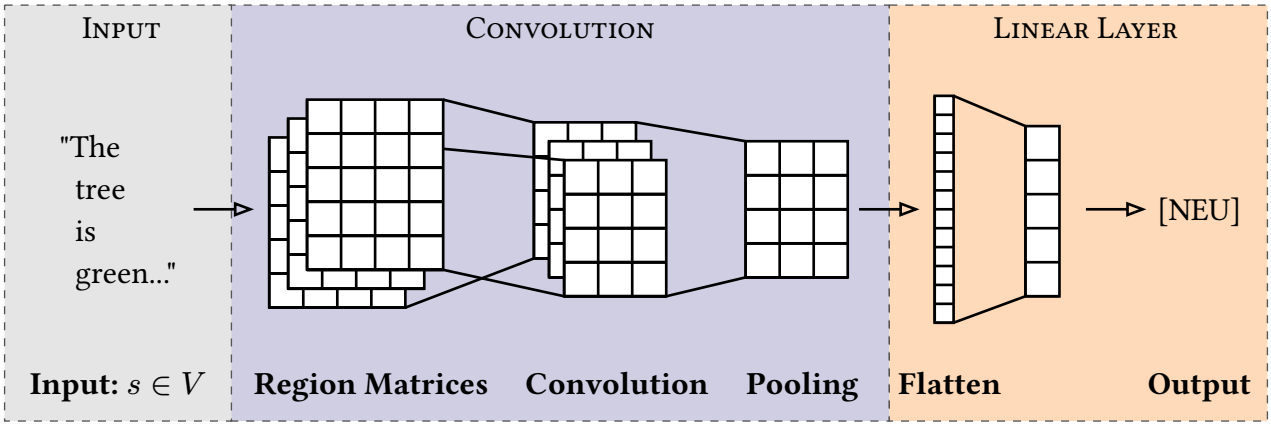


Figure 2.6.: Simple Comvolution based Neural Network to predict an emotion from a given subset of sentences

Figure 2.6 shows such network with the convolutional part. In common architectures multiple convolution and pooling layers are stacked together with an activation function like ReLu in between of each block which improves the output by setting constraints to the convolution outputs. The constraint which is effected by ReLu is the prohibition of negative values by the function $f(x) = \max(0, x)$. This subsequency of the convolution blocks leads to a hierarchical decomposition of the given embedded region matrix. As a result multiple features can be found from each block and each convolution kernel which might result in an higher accuracy. At the end an ordinary decoder in form of a simple linear layer with an activation function like SoftMax is utilized to calculate the resulting emotion from the pooled and flattened vector. Normalization like layer norm or batch norm between each layer can help to ensure better propagation and reduce numerical instability [18].

Another optimizations to fundamental neural network architectures are gating mechanisms from reccurent networks and the uni-directional flow in the model forwarding architectures. They introduce new types of storage cells which allows to store long term features [19]. In the context of emotion detection in text, it might store important context from sentences and words before to differ between fill-words like “this”, “with” and emotionally loaded words. Commonly used are long short term memory cells (LSTM) [20] and gated recurrent units (GRU) [21] which both try to prevent gradient vanishing which is a big problem in recurrent networks. It occurs while propagating backwards through time. Due to many multiplication step, the gradient converges to 0. These two architectures differ mostly in their structure than in their functionality. Additionaly to the GRU cell, LSTM has two more control gates which gives more possibilitys to constraint and rule the memory inside the structure. In both cases they can improve learning features from context and time series.

Transormers are an deep-learning architecture which is based on multi-head attention.

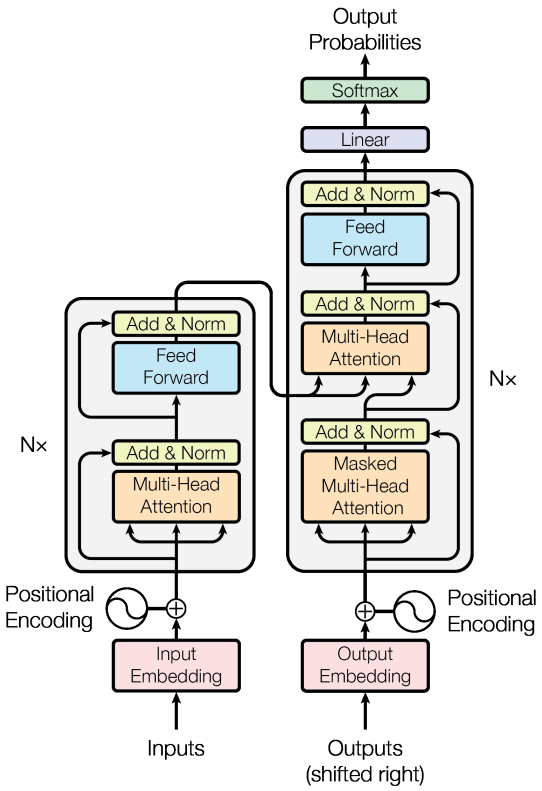


Figure 2.7.: Common transformer model architecture with multi-head attention [22].

The basic idea of attention relies on a huge bottleneck in common neural network architectures because Fixed input vector sizes may result in a large overhead. In the example of natural language processing could this be an result from very large sentences. Attention mechanisms try to solve this issue by focusing the input on to specific parts e.g. training the input weights on important words or subsets from the given sentence.

The resulting transformers are currently setting a milestone in natural language processing and providing the fundaments for ChatGPT from OpenAI [23] or other text understanding models, because of their scalability due to the attention mechanisms and their relieable results. Figure 2.7 shows such a transformer architecture. An input sentence will be feed into the model where it first gets positional encoded thus the position and the frequency for every word in the sentence is stored.

Now the input vectors passes through the first multi-head attention layer to lay the focus on important parts of the input. Neverless the actual input and the weighted part are added and normed together which results in the original sentence but with tagged highlights due to the attention. To learn the features for the output token, the input traverses through a feed forward network which consists of normal linear layers. Likewise the input and the output are matched together with a residual connection.

In addition to the approaches with neural network architectures like transfors, other supervised methods like support vector machines, decision trees or naive bayes are used. These aren't covered in this thesis and relay to future work.

Unsupervised Approaches

2.2.2. Classification in Audio

2.2.3. Classification in Images and Videos

2.3. Datasets

For the multimodal analysis and the supervised approaches in learning a dataset consisting of at minimum video with audio is needed. From this state the text transcription can be retrieved during preprocessing. During this thesis, two different datasets are used which fulfill the requierments. The following datasets are based on the Ekman emotion model with some changes.

CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) [24] consists of 7.442 clips from 91 different actors with their facial and vocal expression. It covers every of Ekmans emotions: *Anger (ANG), Disgust (DIS),Fear (FEA), Happy (HAP), Neutral (NEU), Sad (SAD)*

with the exception of *surprise (SUP)* , because it may can relate to any other emotion with a rapid onset. Additionaly there are a few annotation to the intensity of the sentiment, but it is only a small percentage regarding the amount of data. Therefore it wont't be used for futher evaluation in this thesis.

They recorded twelve different sentence for each type of emotion like "Don't forget a jacket." or "I wonder what this is about." which are semantically neutral connotated. This snippets are between 50 to 80 frames long while it is recorded in 30 frames per second.

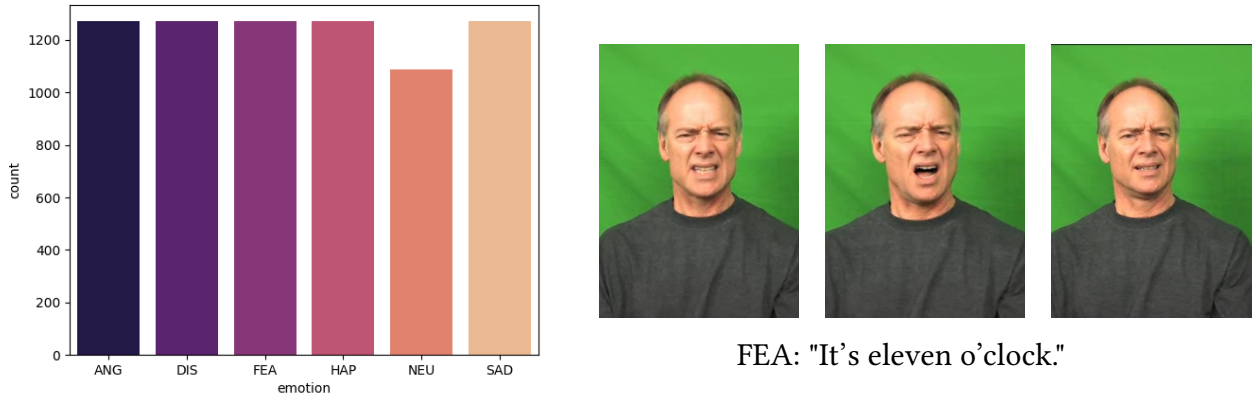
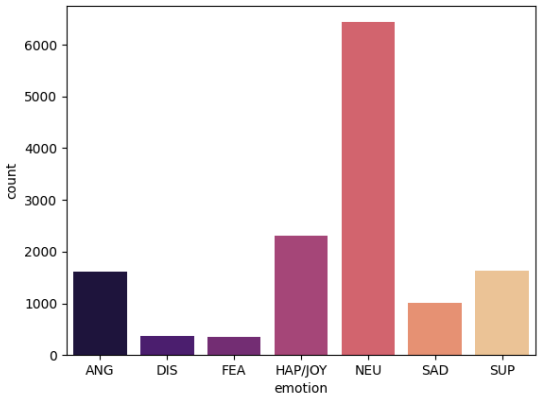


Figure 2.8.: Distribution and example of the emotion presented by the CREMA dataset

As shown in Figure 8 the distribution of labels uniformly distributed with the exception of neutral. This won't leed to any problem, because the actual sentiments shown in the visuell data are very strongly acted so there are clear features available.

MELD (Multimodal EmotionLines Dataset) [25] is a annotated collection of short sequences from the series "Friends". In difference to the CREMA dataset or most other multimodal datasets for sentiment analysis like IEMOCAP [26] the given data has many multi-party conversation where there are more then one participant active and present in the most parts. This leeds to a more complex and challenging feature extraction but approximates more closely the reality.





JOY: "You are a genius." (Chandler)

Figure 2.9.: Distribution and example of the emotion presented by the MELD dataset

Figure 9 shows the uneven distribution of the different sentiments, which can lead to an imbalanced training. Nevertheless it is a quite large dataset with 13.708 annotations which includes in addition to the emotion, the character as the speaking person which can be used to analyize the different features depending on gender or age.

Bibliography

[1] W. James, "What is an Emotion?," *Mind*, vol. 9, no. 34, pp. 188–205, 1884, Accessed: Mar. 11, 2024. [Online]. Available: http://www.jstor.org/stable/2246769

[2] R. W. Picard, "Affective computing." MIT Press, Cambridge, Mass. [u.a.], 1997.

[3] K. R. Scherer and P. Ekman, Eds., *Approaches To Emotion*, 1st ed. Psychology Press, 1984.

[4] P. Kumar and M. Vardhan, "PWEBSA: Twitter sentiment analysis by combining Plutchik wheel of emotion and word embedding," *International Journal of Information Technology*, vol. 14, no. 1, pp. 69–77, Feb. 2022.

[5] D. Chafale and A. Pimpalkar, "Review on Developing Corpora for Sentiment Analysis Using Plutchik's Wheel of Emotions with Fuzzy Logic," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 2, pp. 14–18, 2014.

[6] I. R. Murray and J. L. Arnott, "Applying an analysis of acted vocal emotions to improve the simulation of synthetic speech," *Computer Speech & Language*, vol. 22, no. 2, pp. 107–129, 2008, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0885230807000393

[7] B. Wang, A. Wang, F. Chen, Y. Wang, and C.-C. J. Kuo, "Evaluating word embedding models: methods and experimental results," *APSIPA Transactions on Signal and Information Processing*, vol. 8, p. e19, 2019, doi: 10.1017/ATSIP.2019.12.

[8] D. Karani, "Introduction to word embedding and Word2Vec," *Towards Data Science*, vol. 1, 2018.

[9] Y. Li and T. Yang, "Word Embedding for Understanding Natural Language: A Survey," in *Guide to Big Data Applications*, S. Srinivasan, Ed., Cham: Springer International Publishing, 2018, pp. 83–104.

[10] K. Choudhary and R. Beniwal, "Xplore Word Embedding Using CBOW Model and Skip-Gram Model," in *2021 7th International Conference on Signal Processing and Communication (ICSC)*, 2021, pp. 267–270. doi: 10.1109/ICSC53193.2021.9673321.

[11] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," *arXiv preprint arXiv:1301.3781*, 2013.

[12] L. Ma and Y. Zhang, "Using Word2Vec to process big text data," in *2015 IEEE International Conference on Big Data (Big Data)*, 2015, pp. 2895–2897. doi: 10.1109/BigData.2015.7364114.

[13] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." 2019.

[14] E. Fredkin, "Trie memory," *Commun. ACM*, vol. 3, no. 9, pp. 490–499, Sep. 1960, doi: 10.1145/367390.367400.

[15] X. Song, A. Salcianu, Y. Song, D. Dopson, and D. Zhou, "Fast WordPiece Tokenization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds., Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2089–2103. doi: 10.18653/v1/2021.emnlp-main.160.

[16] C. Fellbaum and K. Brown, "Encyclopedia of language and linguistics." Elsevier Oxford, 2005.

[17] B. Liu, "Text sentiment analysis based on CBOW model and deep learning in big data environment," *Journal of ambient intelligence and humanized computing*, vol. 11, no. 2, pp. 451–458, 2020.

[18] E. S. Lubana, R. Dick, and H. Tanaka, "Beyond BatchNorm: Towards a Unified Understanding of Normalization in Deep Learning," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., Curran Associates, Inc., 2021, pp. 4778–4791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/2578eb9cdf020730f77793e8b58e165a-Paper.pdf

[19] L. Medsker and L. C. Jain, *Recurrent neural networks: design and applications*. CRC press, 1999.

[20] A. Graves, "Long Short-Term Memory," in *Supervised Sequence Labelling with Recurrent Neural Networks*, Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 37–45. doi: 10.1007/978-3-642-24797-2_4.

[21] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling." 2014.

[22] A. Vaswani *et al.*, "Attention Is All You Need." 2023.

[23] OpenAI *et al.*, "GPT-4 Technical Report." 2024.

[24] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.

[25] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," *arXiv preprint arXiv:1810.02508*, 2018.

[26] C. Busso *et al.*, "IEMOCAP: Interactive emotional dyadic motion capture database," *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.

[27] K. Roberts, M. A. Roach, J. Johnson, J. Guthrie, and S. M. Harabagiu, "EmpaTweet: Annotating and Detecting Emotions on Twitter.," in *Lrec*, 2012, pp. 3806–3813.

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Appendix C.

Appendix

Statement of Authorship

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

I am aware of the fact that violations of copyright can lead to injunctive relief and claims for damages of the author as well as a penalty by the law enforcement agency.

Magdeburg, July 31, 2024

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Signature